

THE INFLUENCE OF ROLE PLAYING ON OPINION CHANGE¹

IRVING L. JANIS AND BERT T. KING

Yale University

IN MANY everyday situations, people are induced to play social roles in which they express ideas that are not necessarily in accord with their private convictions. That certain types of role-playing experiences can facilitate changes in personal opinions has been suggested by various impressionistic observations (e.g., Myers [8]). In recent years, psychodramatic techniques which involve role playing have been developed for use in adult education programs, leadership training, employee counseling, and psychotherapy (1, 5, 6, 7, 9). The usual procedure consists of having persons in a group play specified roles in a simulated life situation. One of the main values of this role-playing device, according to its proponents, is that it has a corrective influence on various beliefs and attitudes which underlie chronic difficulties in human relations (cf. Maier [6]).

As yet little is known about the conditions under which role playing leads to actual changes in personal opinions. The present experiment was designed to investigate the effects of one type of demand that is frequently made upon a person when he is induced to play a social role, namely, the requirement that he overtly verbalize to others various opinions which may not correspond to his inner convictions.

As a preliminary step in exploring the effects of role playing, one of the present authors interviewed a group of collegiate debaters who, as members of an organized team, repeatedly were required to play a role in which they publicly expressed views that did not correspond to their personal opinions. Most of the debaters reported that they frequently ended up by accepting the conclusions which

they had been arbitrarily assigned to defend. Myers' (8) impressionistic account of the improvement in morale attitudes produced by participation in an Army public-speaking course points to the same phenomenon and suggests that attitude changes may occur even when role playing is artificially induced. If true, it would appear that "saying is believing"—that overtly expressing an opinion in conformity to social demands will influence the individual's private opinion. Consequently, it seemed worth while to attempt to investigate the effects of this type of role playing in a more controlled laboratory situation where, if the alleged gain from role playing occurs, it might be possible to isolate the critical factors and to explore systematically the mediating mechanisms.

The role-playing effects described above have not as yet been verified by systematic research. If verified, they would still remain open to a variety of alternative explanations. For instance, inducing the individual to play a role in which he must advocate publicly a given position might guarantee exposure to one set of arguments to the exclusion of others. An alternative possibility, however, is that even when exposed to the same persuasive communications, people who are required to verbalize the content to others will tend to be more influenced than those who are only passively exposed. In order to test this hypothesis, the present experiment was designed so that communication exposure would be held constant by comparing the opinion changes of active participants and passive controls who were exposed to the same communications.

METHOD AND PROCEDURES

An initial questionnaire, which was administered as an opinion survey in a large classroom of male college students, contained a series of questions concerning expectations about the future. Included in this "before" questionnaire were the following key opinion items, which dealt with the subject matter of the three communications to which the experimental groups were subsequently exposed:

Item A: During the past year a number of movie

¹ This study was conducted at Yale University as part of a program of research on factors influencing changes in attitude and opinion. The research program is supported by a grant from the Rockefeller Foundation and is under the general direction of Professor Carl I. Hovland, to whom the authors wish to express appreciation for helpful suggestions and criticisms. The authors also wish to thank Professor Fred D. Sheffield for valuable suggestions during discussions preparatory to designing the experiment.

theaters were forced to go out of business as result of television competition and other recent developments. At the present time there are about 18,000 movie theaters remaining. How many commercial movie theaters do you think will be in business three years from now?

Item B: What is your personal estimate about the total supply of meat that will be available for the civilian population of the United States during the year 1953? (. . . — per cent of what it is at present.)

Item C: How many years do you think it will be before a completely effective cure for the common cold is discovered?

The experimental sessions were held approximately four weeks after the initial questionnaire had been filled out, and were represented as being part of a research project designed to develop a new aptitude test for assessing oral speaking ability. The subjects (Ss) were asked to give an informal talk based on an outline prepared by the experimenters (Es) which stated the conclusion and summarized the main arguments to be presented. The arguments were logically relevant but highly biased in that they played up and interpreted "evidence" supporting only one side of the issue. Each active participant was instructed to play the role of a sincere advocate of the given point of view, while two others, who were present at the same experimental session, listened to his talk and read the prepared outline. Each S delivered one of the communications and was passively exposed to the other two. In order to prevent selective attention effects, the active participant was not told what the topic of his talk would be until his turn came to present it. He was given about three minutes to look over the prepared outline, during which time the others (passive controls) also were requested to study duplicate copies of the same outline so as to be prepared for judging the adequacy of the speaker's performance. After the first talk was over, another S was selected to present the second communication, and then the remaining S presented the third communication, the same procedures being followed in each case.

Immediately after the last talk was finished, Ss were given the "after" questionnaire, much of which was devoted to rating the performance of each speaker. The key opinion items were included among numerous filler items, all of which were introduced as questions designed to provide information about the student's interests and opinions concerning the three topics so as to enable the investigators to select the most appropriate topic for future applications of the oral speaking test.

In all three communications, the conclusion specified an opinion estimate which was numerically *lower* than that given by any of the students on the "before" test. Thus, all active participants were required to argue in favor of an extreme position which differed from their initial beliefs. The influence of each communication could readily be observed by noting the degree to which the students in each group *lowered* their opinion estimates on the "after" test.

The basic schema of the experiment is shown in Table 1. In each row of the table which represents exposure to a given communication, there is one group of active participants and two contrasting groups which,

TABLE 1
SCHEMA OF THE EXPERIMENTAL CONDITIONS

	GROUP A (N = 31)	GROUP B (N = 29)	GROUP C (N = 30)
Communication A: movie theaters	active partici- pants	passive controls	passive controls
Communication B: meat supply	passive controls	active partici- pants	passive controls
Communication C: cold cure	passive controls	passive controls	active partici- pants

when combined, form the group of passive controls. In effect, the experimental treatments were repeated with different communication contents, providing three separate instances of active versus passive exposure, although the same Ss were used throughout.

In order to obtain some information for checking on selective attention effects, a variation of the passive control condition (not represented in the table) was introduced into the experiment by using a small supplementary group who listened and took notes on all three talks. In addition, base-line data for assessing the effectiveness of the communications were obtained from a comparable group of "pure" controls who were not exposed to any of the communications.

RESULTS AND DISCUSSION

Effects of active participation. Initially, on each of the three key items in the precommunication questionnaire, the difference between the active participation group and the passive control group was nonsignificant. The opinion changes observed after exposure to the three communications are shown in Table 2.³ The results indicate that in the case of two of the three communications (A and B), the active participants were more influenced than the passive controls. For both communica-

³ The table does not include the data on the "pure" (unexposed) control group. The net changes for this group were approximately zero in the case of all three key items, and the corresponding net changes for the active participants and the passive controls (shown in the last rows of the table) were significantly greater (*p*'s range from .10 down to <.01). Hence, all three communications had a significant effect on the opinions of those who were either actively or passively exposed to them.

The probability values reported throughout this paper are based on one tail of the theoretical distribution. Whenever intergroup comparisons were made with respect to the net percentage who changed by a given amount, the reliability of the difference was tested by the formula presented in Hovland, Lumsdaine, and Sheffield (3, p. 321).

TABLE 2

COMPARISON OF ACTIVE PARTICIPANTS WITH PASSIVE CONTROLS ON AMOUNT OF CHANGE IN OPINION ESTIMATES

CHANGES IN OPINION ESTIMATES†	COMMUNICATION A: (MOVIE THEATERS)		COMMUNICATION B: (MEAT SHORTAGE)		COMMUNICATION C: (COLD CURE)	
	ACTIVE PARTICIPANTS (N = 31)	PASSIVE CONTROLS (N = 57)*	ACTIVE PARTICIPANTS (N = 29)	PASSIVE CONTROLS (N = 57)	ACTIVE PARTICIPANTS (N = 30)	PASSIVE CONTROLS (N = 53)
Sizable increase	0%	2%	0%	2%	7%	6%
Slight increase	3%	9%	7%	14%	10%	9%
No change	23%	20%	24%	16%	13%	19%
Slight decrease	29%	46%	27½%	49%	23%	15%
Sizable decrease	45%	23%	41½%	19%	47%	51%
Total	100%	100%	100%	100%	100%	100%
Net change (% increase minus % decrease)						
Slight or sizable change	-71%	-58%	-62%	-52%	-53%	-51%
Sizable change	-45%	-21%	-41½%	-17%	-40%	-45%
p		.01		.01		> .30

* The number of cases in each passive control group is slightly smaller than expected from the N's shown in Table 1 because the data from a few cases were inadequate and hence were eliminated from the analysis (e.g., the individual failed to give an answer to the particular question).

† The "net change (slight or sizable)" is defined as the percentage changing in the direction advocated by the communication minus the percentage changing in the opposite direction. The "net sizable change" in the case of Communication A refers to the difference in the percentages who lowered and raised their estimate by 5,000 (movie theaters) or more. For Communication B, a sizable change was 25 (per cent) or more; for Communication C it was 5 (years) or more.

tions, the differences in net sizable change are statistically reliable, and the differences in net (slight or sizable) change, although non-reliable, are in the expected direction.

In the case of the third communication (C), the two groups showed approximately the same amount of opinion change. But additional findings (based on confidence ratings given by each S immediately after answering the key opinion questions) indicate that the active participants who presented Communication C, like those who presented the other two communications, expressed a higher level of confidence in their postcommunication estimates than did the corresponding passive controls. Table 3 shows the net changes in confidence ratings for each of the three communications in terms of a breakdown that takes account of the direction and magnitude of opinion change. The breakdown was necessary inasmuch as a successful communication would be expected to increase the confidence only of those who changed their opinions in the direction advocated by the communication. The net change in confidence shown for each subgroup is based on a comparison of pre- and postcommunication ratings given by each S, and was computed by subtracting the per-

centage who showed a decrease in confidence from the percentage who showed an increase in confidence. In general, the findings in Table 3 reveal a consistent pattern for all three communications: in every instance, active participation tended to have at least a slight positive effect with respect to increasing the confidence of those whose opinion estimates were influenced by the communication. The results indicate that active participation resulted in a significant gain in confidence, particularly among those students whose opinion estimates were markedly influenced by Communication C.³ This finding is especially striking in view

³ For the entire group of active participants who were exposed to Communication C, there was a net increase in confidence of 37 per cent; the corresponding net increase for the entire group of passive controls was only 13½ per cent. This difference was due entirely to the marked gain in confidence manifested by those students in the active group who had changed their opinion estimates in the direction advocated by the communication. The results in the first row of the table indicate that, among the students whose opinion estimates were uninfluenced by Communication C, the active participants showed a small net decrease in confidence which was equal to that shown by the passive controls. The next row of Table 3 indicates that, among those students who decreased their opinion estimates by at least one-half year or more after ex-

TABLE 3
COMPARISON OF ACTIVE PARTICIPANTS WITH PASSIVE CONTROLS ON AMOUNT OF CHANGE IN CONFIDENCE

SUBGROUP BREAK- DOWN ACCORDING TO CHANGES IN OPINION ESTIMATES	NET CHANGE IN CONFIDENCE (PER CENT INCREASE MINUS PER CENT DECREASE)					
	COMMUNICATION A		COMMUNICATION B		COMMUNICATION C	
	ACTIVE PARTICIPANTS	PASSIVE CONTROLS	ACTIVE PARTICIPANTS	PASSIVE CONTROLS	ACTIVE PARTICIPANTS	PASSIVE CONTROLS
1. <i>Uninfluenced:</i> opinion estimates increased or un- changed	-12% (N = 8)	-5% (N = 18)	0% (N = 9)	+6% (N = 18)	-11% (N = 9)	-11% (N = 18)
2. <i>Influenced:</i> opinion estimates slightly or siz- ably decreased	+9% (N = 23)	-10% (N = 39)	+20% (N = 20)	+5% (N = 39)	+57% (N = 21)	+26% (N = 35)
Gain from active participation	+19%		+15%		+31%	
3. <i>Highly influenced:</i> opinion estimates sizably decreased	-7% (N = 14)	-38% (N = 13)	+25% (N = 12)	0% (N = 11)	+64½% (N = 14)	+15% (N = 27)
Gain from active participation	+31%		+25%		+49½%	

of the fact that the opinion change results for Communication C (Table 2) failed to show any gain from active participation.

Insofar as confidence ratings can be regarded as indicators of the degree of conviction with which the new opinions are held, the positive findings based on the opinion change data for Communications A and B are partially confirmed by the confidence change data based on Communication C. Thus, the data based on all three communications contribute evidence that the effectiveness of the communications (as manifested by opinion changes or by confidence changes) tended to be augmented by active participation.

Although Ss were not told what their topic would be until they were about to begin giving the talk, it is possible that the ego-involving task of presenting one of the talks may have

posed to Communication C, the active participants showed a greater net increase in confidence than the passive controls; the difference of 31 per cent approaches statistical significance ($p = .07$). Finally, the last row of the table shows that an even greater difference in confidence changes emerges when the comparison is limited to those students who decreased their opinion estimates by five years or more. (The 49½ per cent difference is reliable at beyond the .05 confidence level.) Further analysis of the subgroup data indicated that the differences shown in this table could not be attributed to statistical artifacts arising from initial differences between the various subgroups.

given rise to emotional excitement or other interfering reactions which could have had the effect of reducing the Ss' responsiveness when passively exposed to the other two communications. This possibility appears extremely improbable, however, in the light of supplementary control observations:

1. Some of the passive controls had been exposed to the communications *before* giving their own talk, while others were passively exposed *after* having given their own talk. Nonsignificant differences were found in the amount of opinion change shown under these two conditions.

2. The results from the passive controls were "replicated" by the results from an independent group of 16 students who did not give an oral presentation, but who were asked to follow the prepared outline carefully and to note down the main arguments given by each of the three speakers. Despite the fact that their notes were fairly complete and indicated a relatively high degree of attention to the content of all three communications, these supplementary controls displayed approximately the same amount of opinion change as the original group of passive controls.⁴

⁴ It is conceivable, of course, that the activity of taking notes on the talks might have interfered with

Observations pertinent to explanatory hypotheses. Many different types of speculative hypotheses could be put forth to account for the facilitating effects of active participation, postulating a gain in attention and learning from overtly rehearsing the communication, or a gain in comprehension from reformulating the arguments in one's own words, or a gain in motivation from playing the role of communicator, etc. Some supplementary observations were made for the purpose of exploring various factors which might provide leads to the key mediating mechanisms. Although far from conclusive, the evidence derived from these observations provides a preliminary basis for selecting explanatory hypotheses which warrant further experimental analysis.

The findings based on the supplementary controls (who were required to take notes on the three talks) suggest that variation in attention level probably was not a crucial factor that could explain the participation effects observed in the present experiment. More promising clues were discovered by taking account of differences in the types of reactions evoked by the three communications. We have seen that in the case of Communications A and B, a clear-cut gain from active participation was manifested by changes in opinion estimates; but, in the case of Communication C, opinion estimates were unaffected, the gain being manifested only in the form of increased confidence. With a view to discovering some differentiating factor, we examined the available evidence bearing on the question of why active participation might be more effective under certain stimulus conditions (represented by Communications A and B) than under other conditions (represented by Communication C).

The first step in this inquiry was to examine *E*'s notes on: (a) the active *S*s' behavior while giving their talks, and (b) *S*s' statements in the informal interviews conducted at the end

of each experimental session. These observations provide two suggestive leads:

1. The active participants who presented Communication C seemed to engage in *less improvisation* than those who presented the other two communications. The Communication C group appeared to adhere much more closely to the prepared outline, making little attempt to reformulate the main points, to insert illustrative examples, or to invent additional arguments.

2. Active participants in the Communication C group seemed to experience much more difficulty than the other groups in presenting their talks. During their performance they appeared to be more hesitant and tense. Afterwards, they expressed many more complaints about the task, claiming that their topic was more difficult to present than either of the other two. In general, these subjects seemed *less satisfied* with their performance than those who presented the other two topics.

The first observation suggests that mere repetition of a persuasive communication may have little or no effect as compared with an improvised restatement. This observation is in line with some suggestive findings from an opinion change study by Kelman (4) in which seventh-grade students were given a communication, and, immediately afterwards, were offered various incentives to write essays in support of the communicator's position. Kelman observed that the essays written by the group which showed the greatest amount of opinion change tended to be longer, to include more improvisation, and to be of better over-all quality (as rated by several judges) than the essays written by the other experimental groups.

Reformulating and elaborating on the communication might be a critical factor in producing the gain from active participation, perhaps because the communicatee is stimulated to think of the kinds of arguments, illustrations, and motivating appeals that he regards as most convincing. The importance of the improvisation factor in relation to participation effects could not be investigated further with the data at hand from the present experiment, but is currently being studied by the present authors in another experiment that is specifically designed to compare the effects of different types of active participation.

responsiveness to the persuasive content of the communications. While this possibility cannot be excluded, it seems implausible inasmuch as our *S*s were college students who had had considerable practice in taking notes during lectures. Educational research on the effects of note taking indicates that this form of activity generally has a beneficial rather than a detrimental effect on the student's ability to absorb the content of an oral communication (2).

TABLE 4
SELF-RATINGS OF ACTIVE PARTICIPANTS IN EACH EXPERIMENTAL GROUP

SELF-RATING RESPONSE	EXPERIMENTAL GROUPS (ACTIVE PARTICIPANTS)		
	COMMUNICATION A: (Movie Theaters) (<i>N</i> = 31)	COMMUNICATION B: (Meat Supply) (<i>N</i> = 29)	COMMUNICATION C: (Cold Cure) (<i>N</i> = 30)
1. <i>Over-all performance</i> was at least "satisfactory"	94%	83%	63%
2. Rarely or never spoke in a <i>monotonous</i> tone of voice	64%	76%	53%
3. Rarely or never <i>incoherent</i> in presenting arguments	74%	83%	57%
4. No <i>distortions</i> or <i>misinterpretations</i> of arguments in the prepared outline	32%	52%	13%
5. No <i>omissions</i> of any of the main arguments	74%	72%	70%
6. Succeeded in giving the impression of being " <i>sincere</i> "	52%	52%	43%
Combined rating on all six items:— five or more favorable self-ratings	39%	52%	18%

With respect to the second observation, it should be noted that there may have been an objective basis for the greater dissatisfaction experienced on Communication C because of the greater amount of unfamiliar technical material it contained. The "cold cure" outline referred to a great many technical details concerning the cold virus, antibiotics, allergic reactions, and antihistamines. Many of these details were probably unfamiliar to Ss, and consequently it may have been difficult for them to "spell out" the implications of the arguments. In contrast, the outlines for the other two topics contained very little technical material, relying mainly on arguments that were likely to be quite familiar to college students.

Systematic evidence relevant to Ss' *perception* of the difficulty of presenting each communication was obtained by making use of the self-rating schedule which each student filled out after exposure to the three communications. Table 4 shows the percentage in each experimental group who rated their own performance as adequate or satisfactory on each of six self-appraisal items.

The most comprehensive question was the following: "What is your over-all rating of the informal talk given by this speaker—how good a job do you think he did in presenting his material? — Excellent; — Very Good; — Satisfactory; — Poor; — Very Poor."

The percentage who rated themselves as "satisfactory" or better (shown in the first row of the table) was significantly lower for the group who presented Communication C than for the groups who presented Communications A and B ($p = .002$ and $.04$, respectively). On the remaining five items, each of which dealt with a specific aspect of the speaker's performance, the Communication C group also tended to rate themselves lower than did the other two groups. (On the combined rating, based on all six items, the percentage differences are statistically significant at beyond the .05 confidence level.) The findings consistently indicate that the students in the Communication C group felt less satisfied with their oral speaking performance than did those in the other two groups. Since the group differences in self-ratings tend to parallel the group differences in amount of gain from active participation, the results suggest that *satisfaction with one's own performance* may be a critical factor that determines the magnitude of participation effects.

Further evidence which supports this hypothesis was obtained from an analysis of individual opinion changes, comparing active participants with high and low self-ratings for each of the three communications. For example, among the active participants who presented Communication C, there were 18 students whose self-ratings were comparatively "high" (three to six favorable responses)

and 12 cases whose self-ratings were predominantly "low" (zero, one, or two favorable responses); 55 per cent of the "highs" as against only 17 per cent of the "lows" showed a sizable net opinion change in the direction advocated by the communications ($p = .05$). In general, the comparisons based on all three communications consistently indicate that a greater amount of opinion change occurred among those active participants who rated their oral speaking performance as satisfactory or better. Active participants who felt that they performed poorly, on the other hand, failed to show any more opinion change than the passive controls, and, in the case of Communication C, showed markedly less change than the passive controls ($p = .07$).

During the experimental sessions there were no apparent sources of external social rewards from the environment. Since the others present remained silent, the active participant had no opportunity to know how they were reacting to his talk, except possibly by subtle signs from their facial expressions or from their bodily movements. But even in the absence of any external cues to social approval, it seems probable that *anticipations* concerning such approval would occur if the individual felt that he was performing well, as expressed in his self-ratings. Thus, expectations of favorable audience reactions may have occurred less frequently among Ss who were required to perform the relatively difficult task of presenting the unfamiliar technical material in Communication C than among those who were required to perform the less difficult task of presenting Communication A or B. The increase in opinion change produced by role playing might be mediated by the individual's sense of achievement or his elated feelings about the adequacy of his oral performance. One hypothesis that would follow from this assumption is that when a person conforms outwardly to social demands by playing a role which requires him to advocate a given opinion, he will begin to believe what he is saying if he is made to feel that he says it well.

Although the above hypothesis is suggested by the supplementary correlational findings, it will obviously remain open to question until tested by more precise methods. One cannot be certain that the responses used to assess "satisfaction" represent a separate variable

which is causally related to opinion changes. Acceptance of the communication might be a common factor which inclines those who are most influenced to perceive themselves as having performed well, in which case the self-ratings might merely reflect the same thing as the measures of opinion change. Moreover, even if the two variables can be varied and measured independently, the possibility remains that the observed relationship may be due to some third variable, such as amount of improvisation.

As was noted earlier, the group of active participants who showed the least amount of opinion change not only expressed a low degree of satisfaction but also displayed a relative absence of improvisation in their oral performances. Either the "satisfaction" factor or the "improvisation" factor might prove to be a critical mediating variable. Before drawing a definite conclusion, it is necessary to investigate each factor experimentally—for instance, by giving the Ss "expert" performance ratings which raise or lower their feelings of satisfaction, and by using instructions which increase or decrease the amount of improvisation. These methods are currently being employed in our further research on the effects of role playing.

There is another important problem which arises from the findings in the present experiment and which also requires systematic investigation: Does social role playing facilitate the internalization of externally imposed value judgments, mores, and taboos? The persuasive communications used in this study dealt with relatively impersonal beliefs about the future, and the main findings show that acceptance of opinions of this sort was markedly increased by experimentally induced role playing. It remains problematical, however, whether active participation also influences the acceptance of opinions and attitudes that are more directly tied up with daily life activities, interpersonal relationships, and emotionally charged dilemmas.

Obviously, it is unsafe to generalize widely from a single exploratory study based on the opinion changes of college students produced in a somewhat artificial test situation. Nevertheless, the present experiment provides preliminary evidence indicating that verbal conformity elicited by role playing can significantly influence the acceptance of new

beliefs. Under certain specifiable conditions which await further investigation, it seems to be true that "saying is believing."

SUMMARY AND CONCLUSIONS

The experiment was designed to determine whether or not overt verbalization, induced by role playing, facilitates opinion change. Male college students were assigned at random to two main experimental groups: (a) active participants, who, with the aid of a prepared outline, played the role of a sincere advocate of the given point of view, and (b) passive controls, who silently read and listened to the same communication. In the experimental sessions, three different communications were used, each of which argued in favor of a specific conclusion concerning expected future events and was presented by a different active participant. Opinion measures obtained at the end of the session were compared with the "before" measures obtained about one month earlier.

In general, the active participants tended to be more influenced by the communications than were the passive controls. In the case of two of the communications the active participants showed significantly more opinion change than the passive controls. In the case of the third communication, both groups showed approximately the same amount of opinion change, but active participation, nevertheless, tended to increase the level of confidence of those whose opinion estimates were influenced by the communication. The main findings, together with various methodological checks, support the hypothesis that overt verbalization induced by role playing tends to augment the effectiveness of a persuasive communication.

Additional observations were analyzed in order to explore possible mediating factors underlying the gain in opinion change due to active participation. From behavioral records and interviews, two suggestive leads emerged. In those cases where role playing produced a

marked increase in opinion change: (a) the individual displayed a relatively great amount of improvisation in his talk, and (b) he felt comparatively well satisfied with his oral speaking performance. The first factor suggests that the gain from role playing may occur primarily because the active participant tends to be impressed by his own cogent arguments, clarifying illustrations, and convincing appeals which he is stimulated to think up in order to do a good job of "selling" the idea to others. The second factor suggests an alternative explanation in terms of the rewarding effects of the individual's sense of achievement or feelings of satisfaction with his performance in the role of active participant. Additional evidence pertinent to the second factor, based on a self-rating questionnaire which the Ss filled out immediately after giving the talk, consistently indicated that the greatest amount of opinion change occurred among those active participants who felt that their oral speaking performance was satisfactory. Both the "improvisation" factor and the "satisfaction" factor warrant further investigation.

REFERENCES

1. BAVELAS, A. Role-playing and management training. *Sociatry*, 1947, 1, 183-191.
2. CRAWFORD, C. E. Some experimental studies of the results of college note taking. *J. educ. Res.*, 1925, 12, 379-386.
3. HOVLAND, C. I., LUMSDAINE, A. A., & SHEFFIELD, F. D. *Experiments on mass communication*. Princeton: Princeton Univer. Press, 1949.
4. KELMAN, H. C. Attitude change as a function of response restriction. *Hum. Relat.*, in press.
5. LIPPITT, R. The psychodrama in leadership training. *Sociometry*, 1943, 6, 286-292.
6. MAIER, N. R. F. *Principles of human relations*. New York: Wiley, 1952.
7. MORENO, J. L. *Psychodrama*. Vol. 1. New York: Beacon House, 1946.
8. MYERS, G. C. Control of conduct by suggestion: an experiment in americanization. *J. appl. Psychol.*, 1921, 5, 26-31.
9. ZANDER, A., & LIPPITT, R. Reality-practice as educational method. *Sociometry*, 1944, 7, 129-151.

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